



TEST AUTOMATION ACCELERATORS



.NET TEST AUTOMATION FRAMEWORK

FUNDAMENTAL APPROACH

STANDARDIZED SOLUTIONS
FOR 40%-60% OF
STANDARD/SAME USE CASES

VS

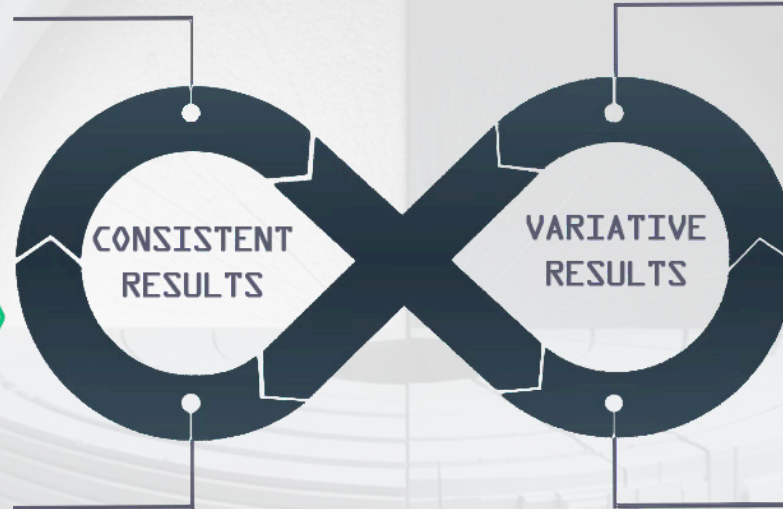
NON-STANDARD SOLUTIONS
FOR 60%-40% OF
NON-STANDARD/SPECIFIC USE CASES

COSTS SAVINGS on AI Usage

HIGHER COSTS on AI Usage

PROCEDURAL
and/or
AI LLM Output VALIDATOR-BASED

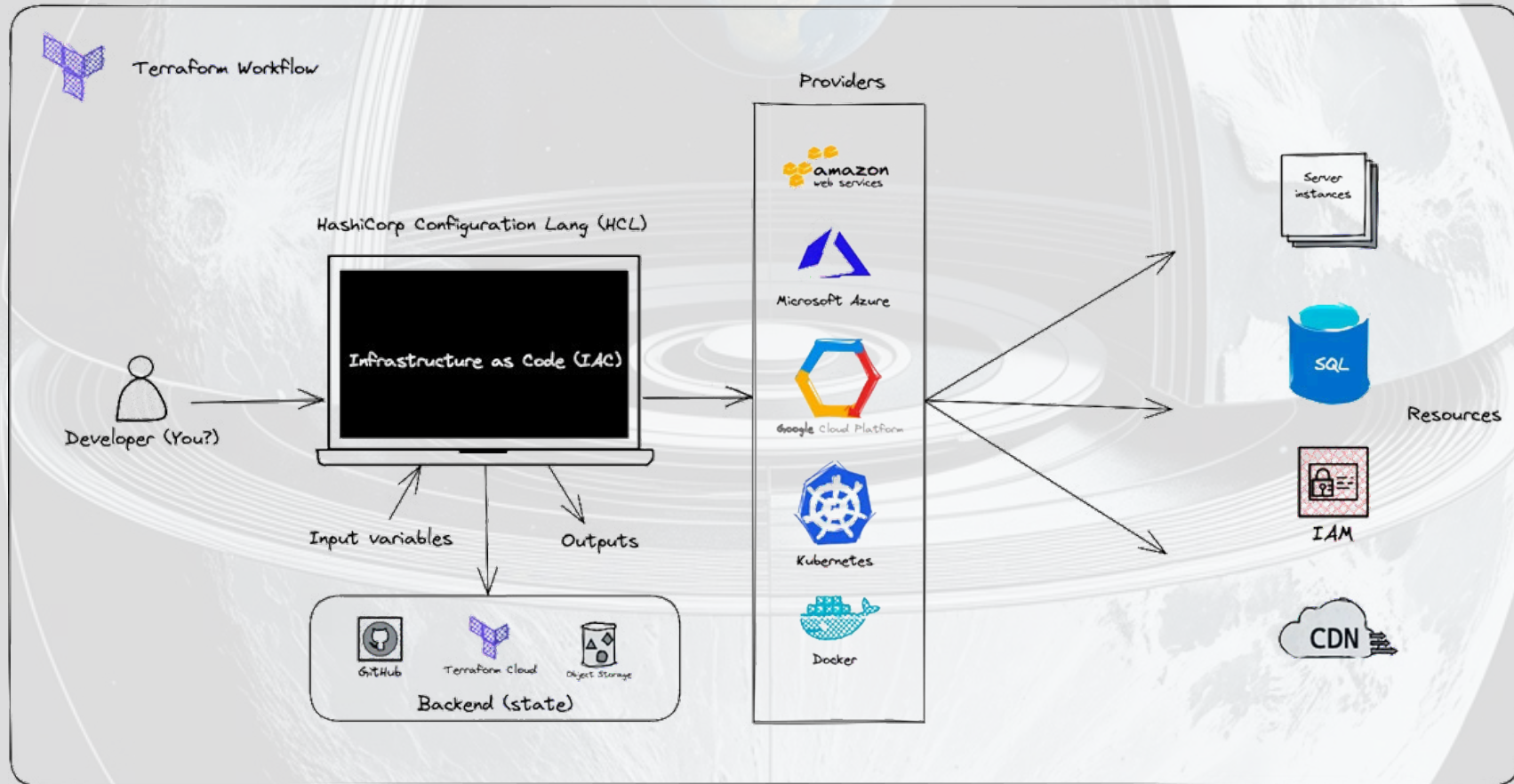
HIGHER PROCESSING SPEEDS
HIGHER RESPONSE TIME



AI CREATIVITY-ORIENTED
with
Elevated Temperature Settings

LOWER PROCESSING SPEEDS
LOWER RESPONSE TIME

SAME EXISTING APPROACH: TERRAFORM





MULTI-CLOUD-PURPOSE
(works with AWS, Azure, GCP, on-prem, etc.)



SCOPE

SINGLE-CLOUD-NATIVE
(AWS CloudFormation GCP Deployment Manager
Azure Resource Manager)



SAME Code - DIFFERENT Providers
&
COMMON LANGUAGE
HashiCorp Configuration Language



PORTABILITY & LANGUAGE



DIFFERENT Providers - DIFFERENT Code
&
DIFFERENT LANGUAGES
(SINGLE-CLOUD NATIVE Languages)



LEARN SINGLE SYNTAX
for ALL Clouds



LEARNING CURVE

LEARN DIFFERENT SYNTAX
for DIFFERENT PROVIDERS



LARGE ECOSYSTEM of REUSABLE MODULES
Across Providers



COMMUNITY & MODULES



MODULES/Templates
LIMITED to SINGLE CLOUD





MULTI-PROJECT-PURPOSE



SCOPE

SINGLE-PROJECT-NATIVE



SAME/SIMILAR Code – DIFFERENT Projects
&
COMMON CONVENTIONS
(within both Same
and across Different Languages)



PORTABILITY & LANGUAGE

DIFFERENT Projects – DIFFERENT Code
&
DIFFERENT CONVENTIONS
(within both Same
and across Different Languages)



LEARN SINGLE SYNTAX
for ALL Projects and Languages



LEARNING CURVE

LEARN DIFFERENT SYNTAX
for DIFFERENT Projects and Languages



LARGE ECOSYSTEM of REUSABLE MODULES
Across Projects and Languages



COMMUNITY & MODULES

MODULES/Templates
LIMITED to Single Project and Language



SHARED APPROACH ADVANTAGES



BEST of the **BEST PRACTICES** from Test Automation Frameworks and Architects across All Projects
GATHERED TOGETHER in **SINGLE SHARED APPROACH** Implementations



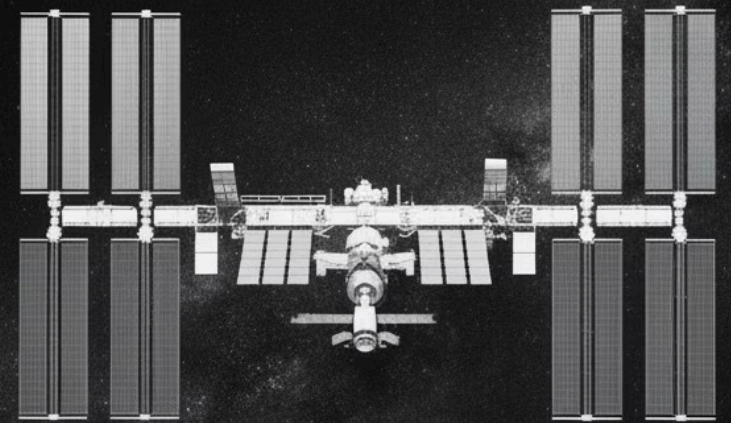
SWITCH between **PROJECTS** with the **SAME** Programming Language with **MINIMUM LEARNING CURVE**



SWITCH between **PROJECTS** with **DIFFERENT** Programming Languages with **MINIMIZED LEARNING CURVE**



Some **PACKAGES** can be **SHARED** between Developers and Test Automation Engineers



NUGETS + LINK

A detailed illustration of a futuristic space station interior. A large, octagonal window dominates the center, offering a view of a barren, cratered planet surface in the foreground and a distant, blue and white planet (Earth) in the upper right against a starry space background. The station's interior is composed of light gray, metallic-looking panels with various vents, dials, and structural details. The lighting is soft and even, highlighting the textures of the station and the landscape outside.

UI



ACCELERATOR STATE

TECHNOLOGY-AGNOSTIC

DIFFERENT TECHNOLOGIES - SAME CODEBASE



Selenium

and



Playwright

CHANGE/ADD NEW TECHNOLOGY

SIMPLE SWITCH

NO NEED TO REWRITE TESTS' CODE

```
private void AddUiInteractorTechnology()
{
    if (CurrentTechnology == UiInteractorTechnology.Playwright)
    {
        _serviceCollection.ConfigurePlaywright(
            customBrowserFactoryFunc: (provider) => { return new PlaywrightBrowserFactory(provider.GetRequiredService<ILogger>()); },
            getCurrentTestIdentifierFunc: () => TestContext.CurrentContext.Test.ID,
            getCurrentFeatureIdentifierFunc: () => TestContext.CurrentContext.Test.ClassName);
    }
    else if (CurrentTechnology == UiInteractorTechnology.Selenium)
    {
        _serviceCollection.ConfigureSelenium(
            customWebDriverFactoryFunc: (provider) => { return new SeleniumWebDriverFactory(provider.GetRequiredService<ILogger>()); },
            getCurrentTestIdentifierFunc: () => TestContext.CurrentContext.Test.ID,
            getCurrentFeatureIdentifierFunc: () => TestContext.CurrentContext.Test.ClassName);
    }
}
```



BASE/DEFAULT STATE

TECHNOLOGY-SPECIFIC

DIFFERENT TECHNOLOGIES - DIFFERENT CODEBASE



Selenium

or



Playwright

CHANGE/ADD NEW TECHNOLOGY

'HARD RESET'

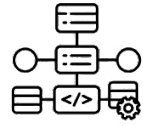
REWRITE OF HARDCODED TECHNOLOGY TESTS' CODE

IS REQUIRED

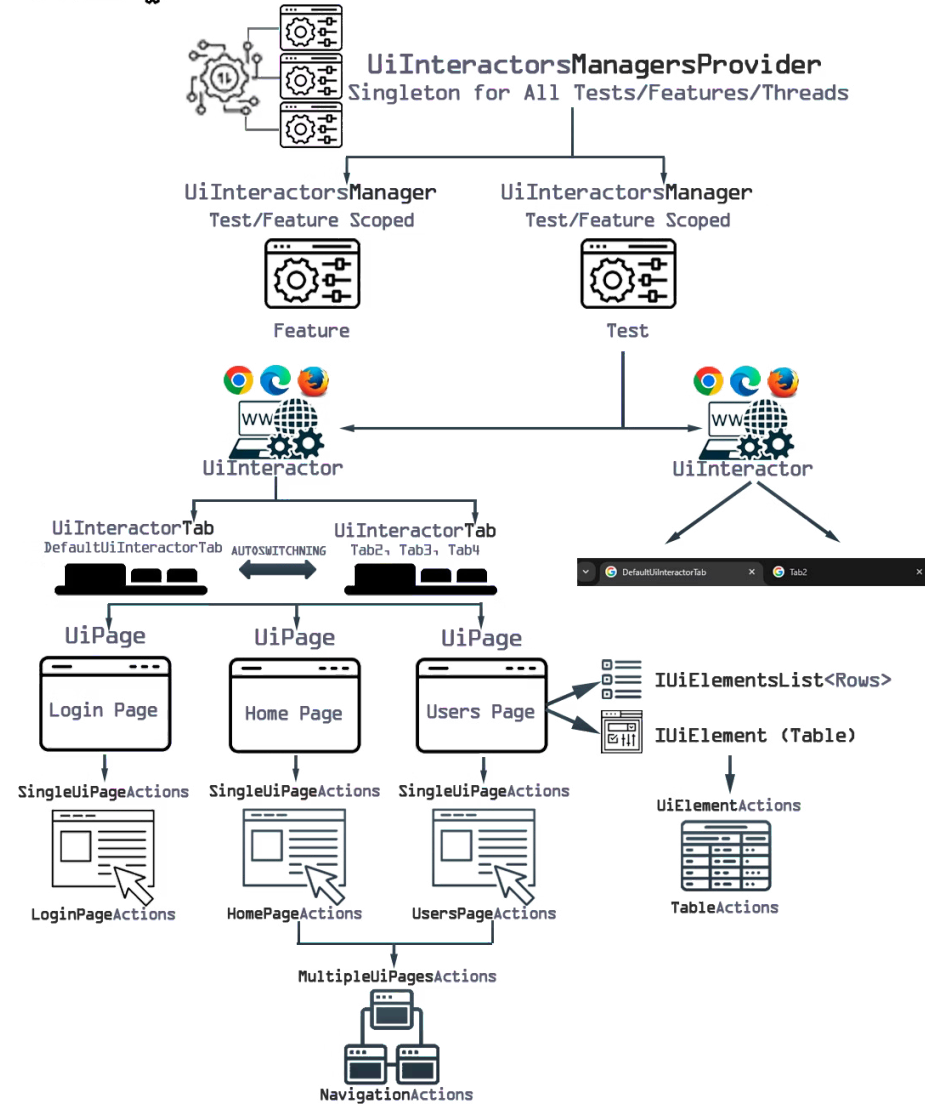
```
var shadowHost = _driver.FindElement(By.CssSelector("#shadow_host"));
var shadowRoot = shadowHost.GetShadowRoot();
var shadowContent = shadowRoot.FindElement(By.CssSelector("#shadow_content"));

var locator = Page.GetByRole(AriaRole.Button, new() { Name = "Sign in" });

await locator.HoverAsync();
await locator.ClickAsync();
```



GENERAL ARCHITECTURE





ACCELERATOR STATE (WITH AUTOSWITCHING)

SUPPORTED UIELEMENTS SEARCH PAIRWISE COMBINATIONS

I - RELATIVE TO THE BEGINNING OF PAGE

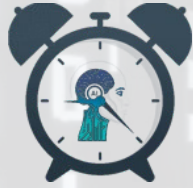
- 1. Page -> Simple_UIElement_AsInterface
- 2. Page -> Complex_UIElement_AsClass
- 3. Page -> UiElementsList_Of_Simple_UIElements_AsInterface
- 4. Page -> UiElementsList_Of_Complex_UIElements_AsClass
- 5. Page -> Frame_Simple_UIElement_AsInterface
- 6. Page -> UiElementsList_Of_Frame_SimpleUIElements_AsInterface
- 7. Page -> Frame_Complex_UIElement_AsClass -> Frame_Simple_UIElement_AsInterface
- 8. Page -> ShadowRootHost_Simple_UIElement_AsInterface
- 9. Page -> UiElementsList_Of_ShadowRootHost_SimpleUIElements_AsInterface
- 10. Page -> ShadowRootHost_Complex_UIElement_AsClass -> ShadowRootHost_Simple_UIElement_AsInterface

II - RELATIVE TO COMPLEX UIELEMENT AS CLASS

- 11. Page -> Complex_UIElement_AsClass -> Simple_UIElement_AsInterface
- 12. Page -> Complex_UIElement_AsClass -> Complex_UIElement_AsClass
- 13. Page -> Complex_UIElement_AsClass -> UiElementsList_Of_Simple_UIElements_AsInterface
- 14. Page -> Complex_UIElement_AsClass -> UiElementsList_Of_Complex_UIElements_AsClass
- 15. Page -> Complex_UIElement_AsClass -> Frame_Simple_UIElement_AsInterface
- 16. Page -> Complex_UIElement_AsClass -> UiElementsList_Of_Frame_SimpleUIElements_AsInterface
- 17. Page -> Complex_UIElement_AsClass -> Frame_Complex_UIElement_AsClass -> Frame_Simple_UIElement_AsInterface
- 18. Page -> Complex_UIElement_AsClass -> ShadowRootHost_Simple_UIElement_AsInterface
- 19. Page -> Complex_UIElement_AsClass -> UiElementsList_Of_ShadowRootHost_Simple_UIElements_AsInterface
- 20. Page -> Complex_UIElement_AsClass -> ShadowRootHost_Complex_UIElement_AsClass -> ShadowRootHost_Simple_UIElement_AsInterface

III - RELATIVE TO UIELEMENTS LIST OF COMPLEX UIELEMENTS AS CLASS

- 21. Page -> UiElementsList_Of_Complex_UIElements_AsClass -> Simple_UIElement_AsInterface
- 22. Page -> UiElementsList_Of_Complex_UIElements_AsClass -> Complex_UIElement_AsClass
- 23. Page -> UiElementsList_Of_Complex_UIElements_AsClass -> UiElementsList_Of_Simple_UIElements_AsInterface
- 24. Page -> UiElementsList_Of_Complex_UIElements_AsClass -> UiElementsList_Of_Complex_UIElements_AsClass
- 25. Page -> UiElementsList_Of_Complex_UIElements_AsClass -> Frame_Simple_UIElement_AsInterface
- 26. Page -> UiElementsList_Of_Complex_UIElements_AsClass -> UiElementsList_Of_Frame_SimpleUIElements_AsInterface
- 27. Page -> UiElementsList_Of_Complex_UIElements_AsClass -> Frame_Complex_UIElement_AsClass -> Frame_Simple_UIElement_AsInterface
- 28. Page -> UiElementsList_Of_Complex_UIElements_AsClass -> ShadowRootHost_Simple_UIElement_AsInterface
- 29. Page -> UiElementsList_Of_Complex_UIElements_AsClass -> UiElementsList_Of_ShadowRootHost_Simple_UIElements_AsInterface
- 30. Page -> UiElementsList_Of_Complex_UIElements_AsClass -> ShadowRootHost_Complex_UIElement_AsClass -> ShadowRootHost_Simple_UIElement_AsInterface



ACCELERATOR STATE (WITH AUTOSWITCHING)

SUPPORTED UIELEMENTS SEARCH PAIRWISE COMBINATIONS

IV – RELATIVE TO FRAME COMPLEX UIELEMENT AS CLASS

- 31. Page -> Frame_Complex_UIElement_AsClass -> Simple_UIElement_AsInterface
- 32. Page -> Frame_Complex_UIElement_AsClass -> Complex_UIElement_AsClass
- 33. Page -> Frame_Complex_UIElement_AsClass -> UIElementsList_Of_Simple_UIElements_AsInterface
- 34. Page -> Frame_Complex_UIElement_AsClass -> UIElementsList_Of_Complex_UIElements_AsClass
- 35. Page -> Frame_Complex_UIElement_AsClass -> Frame_Simple_UIElement_AsInterface
- 36. Page -> Frame_Complex_UIElement_AsClass -> UIElementsList_Of_Frame_Simple_UIElements_AsInterface
- 37. Page -> Frame_Complex_UIElement_AsClass -> Frame_Complex_UIElement_AsClass -> Frame_Simple_UIElement_AsInterface
- 38. Page -> Frame_Complex_UIElement_AsClass -> ShadowRootHost_Simple_UIElement_AsInterface
- 39. Page -> Frame_Complex_UIElement_AsClass -> UIElementsList_Of_ShadowRootHost_Simple_UIElements_AsInterface
- 40. Page -> Frame_Complex_UIElement_AsClass -> ShadowRootHost_Complex_UIElement_AsClass -> ShadowRootHost_Simple_UIElement_AsInterface

V – RELATIVE TO UIELEMENTS LIST OF FRAME COMPLEX UIELEMENTS AS CLASS

- 41. Page -> UIElementsList_Of_Frame_Complex_UIElements_AsClass -> Simple_UIElement_AsInterface
- 42. Page -> UIElementsList_Of_Frame_Complex_UIElements_AsClass -> Complex_UIElement_AsClass
- 43. Page -> UIElementsList_Of_Frame_Complex_UIElements_AsClass -> UIElementsList_Of_Simple_UIElements_AsInterface
- 44. Page -> UIElementsList_Of_Frame_Complex_UIElements_AsClass -> UIElementsList_Of_Complex_UIElements_AsClass
- 45. Page -> UIElementsList_Of_Frame_Complex_UIElements_AsClass -> Frame_Simple_UIElement_AsInterface
- 46. Page -> UIElementsList_Of_Frame_Complex_UIElements_AsClass -> UIElementsList_Of_Frame_Simple_UIElements_AsInterface
- 47. Page -> UIElementsList_Of_Frame_Complex_UIElements_AsClass -> Frame_Complex_UIElement_AsClass -> Frame_Simple_UIElement_AsInterface
- 48. Page -> UIElementsList_Of_Frame_Complex_UIElements_AsClass -> ShadowRootHost_Simple_UIElement_AsInterface
- 49. Page -> UIElementsList_Of_Frame_Complex_UIElements_AsClass -> UIElementsList_Of_ShadowRootHost_Simple_UIElements_AsInterface
- 50. Page -> UIElementsList_Of_Frame_Complex_UIElements_AsClass -> ShadowRootHost_Complex_UIElement_AsClass -> ShadowRootHost_Simple_UIElement_AsInterface

VI – RELATIVE TO 2 FRAMES COMPLEX UIELEMENTS AS CLASS

- 51. Page -> Frame_Complex_UIElement_AsClass -> Frame_Complex_UIElement_AsClass -> Simple_UIElement_AsInterface
- 52. Page -> Frame_Complex_UIElement_AsClass -> Frame_Complex_UIElement_AsClass -> Complex_UIElement_AsClass
- 53. Page -> Frame_Complex_UIElement_AsClass -> Frame_Complex_UIElement_AsClass -> UIElementsList_Of_Simple_UIElements_AsInterface
- 54. Page -> Frame_Complex_UIElement_AsClass -> Frame_Complex_UIElement_AsClass -> UIElementsList_Of_Complex_UIElements_AsClass
- 55. Page -> Frame_Complex_UIElement_AsClass -> Frame_Complex_UIElement_AsClass -> UIElementsList_Of_Frame_Simple_UIElements_AsInterface
- 56. Page -> Frame_Complex_UIElement_AsClass -> Frame_Complex_UIElement_AsClass -> ShadowRootHost_Simple_UIElement_AsInterface
- 57. Page -> Frame_Complex_UIElement_AsClass -> Frame_Complex_UIElement_AsClass -> UIElementsList_Of_ShadowRootHost_Simple_UIElements_AsInterface
- 58. Page -> Frame_Complex_UIElement_AsClass -> Frame_Complex_UIElement_AsClass -> ShadowRootHost_Complex_UIElement_AsClass -> ShadowRootHost_Simple_UIElement_AsInterface



ACCELERATOR STATE (WITH AUTOSWITCHING)

SUPPORTED UIELEMENTS SEARCH PAIRWISE COMBINATIONS

VII - RELATIVE TO SHADOW ROOT HOST COMPLEX UIELEMENT AS CLASS

- ✓ 59. Page -> ShadowRootHost_Complex_UIElement_AsClass -> Simple_UIElement_AsInterface
- ✓ 60. Page -> ShadowRootHost_Complex_UIElement_AsClass -> Complex_UIElement_AsClass
- ✓ 61. Page -> ShadowRootHost_Complex_UIElement_AsClass -> UIElementsList_Of_Simple_UIElements_AsInterface
- ✓ 62. Page -> ShadowRootHost_Complex_UIElement_AsClass -> UIElementsList_Of_Complex_UIElements_AsClass
- ✓ 63. Page -> ShadowRootHost_Complex_UIElement_AsClass -> Frame_Simple_UIElement_AsInterface
- ✓ 64. Page -> ShadowRootHost_Complex_UIElement_AsClass -> UIElementsList_Of_Frame_Simple_UIElements_AsInterface
- ✓ 65. Page -> ShadowRootHost_Complex_UIElement_AsClass -> Frame_Complex_UIElement_AsClass -> Frame_Simple_UIElement_AsInterface
- ✓ 66. Page -> ShadowRootHost_Complex_UIElement_AsClass -> ShadowRootHost_Simple_UIElement_AsInterface
- ✓ 67. Page -> ShadowRootHost_Complex_UIElement_AsClass -> UIElementsList_Of_ShadowRootHost_Simple_UIElements_AsInterface
- ✓ 68. Page -> ShadowRootHost_Complex_UIElement_AsClass -> ShadowRootHost_Complex_UIElement_AsClass -> ShadowRootHost_Simple_UIElement_AsInterface

VIII - RELATIVE TO UIELEMENTS LIST OF SHADOW ROOT HOST COMPLEX UIELEMENTS AS CLASS

- ✓ 69. Page -> UIElementsList_Of_ShadowRootHost_Complex_UIElements_AsClass -> Simple_UIElement_AsInterface
- ✓ 70. Page -> UIElementsList_Of_ShadowRootHost_Complex_UIElements_AsClass -> Complex_UIElement_AsClass
- ✓ 71. Page -> UIElementsList_Of_ShadowRootHost_Complex_UIElements_AsClass -> UIElementsList_Of_Simple_UIElements_AsInterface
- ✓ 72. Page -> UIElementsList_Of_ShadowRootHost_Complex_UIElements_AsClass -> UIElementsList_Of_Complex_UIElements_AsClass
- ✓ 73. Page -> UIElementsList_Of_ShadowRootHost_Complex_UIElements_AsClass -> Frame_Simple_UIElement_AsInterface
- ✓ 74. Page -> UIElementsList_Of_ShadowRootHost_Complex_UIElements_AsClass -> UIElementsList_Of_Frame_Simple_UIElements_AsInterface
- ✓ 75. Page -> UIElementsList_Of_ShadowRootHost_Complex_UIElements_AsClass -> Frame_Complex_UIElement_AsClass -> Frame_Simple_UIElement_AsInterface
- ✓ 76. Page -> UIElementsList_Of_ShadowRootHost_Complex_UIElements_AsClass -> ShadowRootHost_Simple_UIElement_AsInterface
- ✓ 77. Page -> UIElementsList_Of_ShadowRootHost_Complex_UIElements_AsClass -> UIElementsList_Of_ShadowRootHost_Simple_UIElements_AsInterface
- ✓ 78. Page -> UIElementsList_Of_ShadowRootHost_Complex_UIElements_AsClass -> ShadowRootHost_Complex_UIElement_AsClass -> ShadowRootHost_Simple_UIElement_AsInterface

IX - RELATIVE TO 2 SHADOW ROOT HOST COMPLEX UIELEMENTS AS CLASS

- ✓ 79. Page -> ShadowRootHost_Complex_UIElement_AsClass -> ShadowRootHost_Complex_UIElement_AsClass -> Simple_UIElement_AsInterface
- ✓ 80. Page -> ShadowRootHost_Complex_UIElement_AsClass -> ShadowRootHost_Complex_UIElement_AsClass -> Complex_UIElement_AsClass
- ✓ 81. Page -> ShadowRootHost_Complex_UIElement_AsClass -> ShadowRootHost_Complex_UIElement_AsClass -> UIElementsList_Of_Simple_UIElements_AsInterface
- ✓ 82. Page -> ShadowRootHost_Complex_UIElement_AsClass -> ShadowRootHost_Complex_UIElement_AsClass -> UIElementsList_Of_Complex_UIElements_AsClass
- ✓ 83. Page -> ShadowRootHost_Complex_UIElement_AsClass -> ShadowRootHost_Complex_UIElement_AsClass -> Frame_Simple_UIElement_AsInterface
- ✓ 84. Page -> ShadowRootHost_Complex_UIElement_AsClass -> ShadowRootHost_Complex_UIElement_AsClass -> UIElementsList_Of_Frame_Simple_UIElements_AsInterface
- ✓ 85. Page -> ShadowRootHost_Complex_UIElement_AsClass -> ShadowRootHost_Complex_UIElement_AsClass -> Frame_Complex_UIElement_AsClass -> Frame_Simple_UIElement_AsInterface
- ✓ 86. Page -> ShadowRootHost_Complex_UIElement_AsClass -> ShadowRootHost_Complex_UIElement_AsClass -> UIElementsList_Of_ShadowRootHost_Simple_UIElements_AsInterface



STATE OF MOST BASE/DEFAULT FRAMEWORKS (WITHOUT AUTOSWITCHING)

SUPPORTED UIELEMENTS SEARCH PAIRWISE COMBINATIONS

I - RELATIVE TO THE BEGINNING OF PAGE

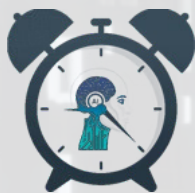
- ✓ 1. Page -> Simple_UIElement_AsInterface
- ✓ 2. Page -> Complex_UIElement_AsClass
- ✓ 3. Page -> UiElementsList_Of_Simple_UIElements_AsInterface
- ✓ 4. Page -> UiElementsList_Of_Complex_UIElements_AsClass
- ✗ 5. Page -> Frame_Simple_UIElement_AsInterface
- ✗ 6. Page -> UiElementsList_Of_Frame_SimpleUIElements_AsInterface
- ✗ 7. Page -> Frame_Complex_UIElement_AsClass -> Frame_Simple_UIElement_AsInterface
- ✗ 8. Page -> ShadowRootHost_Simple_UIElement_AsInterface
- ✗ 9. Page -> UiElementsList_Of_ShadowRootHost_SimpleUIElements_AsInterface
- ✗ 10. Page -> ShadowRootHost_Complex_UIElement_AsClass -> ShadowRootHost_Simple_UIElement_AsInterface

II - RELATIVE TO COMPLEX UIELEMENT AS CLASS

- ✓ 11. Page -> Complex_UIElement_AsClass -> Simple_UIElement_AsInterface
- ✓ 12. Page -> Complex_UIElement_AsClass -> Complex_UIElement_AsClass
- ✓ 13. Page -> Complex_UIElement_AsClass -> UiElementsList_Of_Simple_UIElements_AsInterface
- ✓ 14. Page -> Complex_UIElement_AsClass -> UiElementsList_Of_Complex_UIElements_AsClass
- ✗ 15. Page -> Complex_UIElement_AsClass -> Frame_Simple_UIElement_AsInterface
- ✗ 16. Page -> Complex_UIElement_AsClass -> UiElementsList_Of_Frame_Simple_UIElements_AsInterface
- ✗ 17. Page -> Complex_UIElement_AsClass -> Frame_Complex_UIElement_AsClass -> Frame_Simple_UIElement_AsInterface
- ✗ 18. Page -> Complex_UIElement_AsClass -> ShadowRootHost_Simple_UIElement_AsInterface
- ✗ 19. Page -> Complex_UIElement_AsClass -> UiElementsList_Of_ShadowRootHost_Simple_UIElements_AsInterface
- ✗ 20. Page -> Complex_UIElement_AsClass -> ShadowRootHost_Complex_UIElement_AsClass -> ShadowRootHost_Simple_UIElement_AsInterface

III - RELATIVE TO UIELEMENTS LIST OF COMPLEX UIELEMENTS AS CLASS

- ✓ 21. Page -> UiElementsList_Of_Complex_UIElements_AsClass -> Simple_UIElement_AsInterface
- ✓ 22. Page -> UiElementsList_Of_Complex_UIElements_AsClass -> Complex_UIElement_AsClass
- ✓ 23. Page -> UiElementsList_Of_Complex_UIElements_AsClass -> UiElementsList_Of_Simple_UIElements_AsInterface
- ✓ 24. Page -> UiElementsList_Of_Complex_UIElements_AsClass -> UiElementsList_Of_Complex_UIElements_AsClass
- ✗ 25. Page -> UiElementsList_Of_Complex_UIElements_AsClass -> Frame_Simple_UIElement_AsInterface
- ✗ 26. Page -> UiElementsList_Of_Complex_UIElements_AsClass -> UiElementsList_Of_Frame_Simple_UIElements_AsInterface
- ✗ 27. Page -> UiElementsList_Of_Complex_UIElements_AsClass -> Frame_Complex_UIElement_AsClass -> Frame_Simple_UIElement_AsInterface
- ✗ 28. Page -> UiElementsList_Of_Complex_UIElements_AsClass -> ShadowRootHost_Simple_UIElement_AsInterface
- ✗ 29. Page -> UiElementsList_Of_Complex_UIElements_AsClass -> UiElementsList_Of_ShadowRootHost_Simple_UIElements_AsInterface
- ✗ 30. Page -> UiElementsList_Of_Complex_UIElements_AsClass -> ShadowRootHost_Complex_UIElement_AsClass -> ShadowRootHost_Simple_UIElement_AsInterface



STATE OF MOST BASE/DEFAULT FRAMEWORKS (WITHOUT AUTOSWITCHING)

SUPPORTED UIELEMENTS SEARCH PAIRWISE COMBINATIONS

IV - RELATIVE TO FRAME COMPLEX UIELEMENT AS CLASS

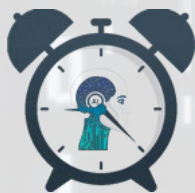
- 31. Page -> Frame_Complex_UIElement_AsClass -> Simple_UIElement_AsInterface
- 32. Page -> Frame_Complex_UIElement_AsClass -> Complex_UIElement_AsClass
- 33. Page -> Frame_Complex_UIElement_AsClass -> UIElementsList_Of_Simple_UIElements_AsInterface
- 34. Page -> Frame_Complex_UIElement_AsClass -> UIElementsList_Of_Complex_UIElements_AsClass
- 35. Page -> Frame_Complex_UIElement_AsClass -> Frame_Simple_UIElement_AsInterface
- 36. Page -> Frame_Complex_UIElement_AsClass -> UIElementsList_Of_Frame_Simple_UIElements_AsInterface
- 37. Page -> Frame_Complex_UIElement_AsClass -> Frame_Complex_UIElement_AsClass -> Frame_Simple_UIElement_AsInterface
- 38. Page -> Frame_Complex_UIElement_AsClass -> ShadowRootHost_Simple_UIElement_AsInterface
- 39. Page -> Frame_Complex_UIElement_AsClass -> UIElementsList_Of_ShadowRootHost_Simple_UIElements_AsInterface
- 40. Page -> Frame_Complex_UIElement_AsClass -> ShadowRootHost_Complex_UIElement_AsClass -> ShadowRootHost_Simple_UIElement_AsInterface

V - RELATIVE TO UIELEMENTS LIST OF FRAME COMPLEX UIELEMENTS AS CLASS

- 41. Page -> UIElementsList_Of_Frame_Complex_UIElements_AsClass -> Simple_UIElement_AsInterface
- 42. Page -> UIElementsList_Of_Frame_Complex_UIElements_AsClass -> Complex_UIElement_AsClass
- 43. Page -> UIElementsList_Of_Frame_Complex_UIElements_AsClass -> UIElementsList_Of_Simple_UIElements_AsInterface
- 44. Page -> UIElementsList_Of_Frame_Complex_UIElements_AsClass -> UIElementsList_Of_Complex_UIElements_AsClass
- 45. Page -> UIElementsList_Of_Frame_Complex_UIElements_AsClass -> Frame_Simple_UIElement_AsInterface
- 46. Page -> UIElementsList_Of_Frame_Complex_UIElements_AsClass -> UIElementsList_Of_Frame_Simple_UIElements_AsInterface
- 47. Page -> UIElementsList_Of_Frame_Complex_UIElements_AsClass -> Frame_Complex_UIElement_AsClass -> Frame_Simple_UIElement_AsInterface
- 48. Page -> UIElementsList_Of_Frame_Complex_UIElements_AsClass -> ShadowRootHost_Simple_UIElement_AsInterface
- 49. Page -> UIElementsList_Of_Frame_Complex_UIElements_AsClass -> UIElementsList_Of_ShadowRootHost_Simple_UIElements_AsInterface
- 50. Page -> UIElementsList_Of_Frame_Complex_UIElements_AsClass -> ShadowRootHost_Complex_UIElement_AsClass -> ShadowRootHost_Simple_UIElement_AsInterface

VI - RELATIVE TO 2 FRAMES COMPLEX UIELEMENTS AS CLASS

- 51. Page -> Frame_Complex_UIElement_AsClass -> Frame_Complex_UIElement_AsClass -> Simple_UIElement_AsInterface
- 52. Page -> Frame_Complex_UIElement_AsClass -> Frame_Complex_UIElement_AsClass -> Complex_UIElement_AsClass
- 53. Page -> Frame_Complex_UIElement_AsClass -> Frame_Complex_UIElement_AsClass -> UIElementsList_Of_Simple_UIElements_AsInterface
- 54. Page -> Frame_Complex_UIElement_AsClass -> Frame_Complex_UIElement_AsClass -> UIElementsList_Of_Complex_UIElements_AsClass
- 55. Page -> Frame_Complex_UIElement_AsClass -> Frame_Complex_UIElement_AsClass -> UIElementsList_Of_Frame_Simple_UIElements_AsInterface
- 56. Page -> Frame_Complex_UIElement_AsClass -> Frame_Complex_UIElement_AsClass -> ShadowRootHost_Simple_UIElement_AsInterface
- 57. Page -> Frame_Complex_UIElement_AsClass -> Frame_Complex_UIElement_AsClass -> UIElementsList_Of_ShadowRootHost_Simple_UIElements_AsInterface
- 58. Page -> Frame_Complex_UIElement_AsClass -> Frame_Complex_UIElement_AsClass -> ShadowRootHost_Complex_UIElement_AsClass -> ShadowRootHost_Simple_UIElement_AsInterface



STATE OF MOST BASE/DEFAULT FRAMEWORKS (WITHOUT AUTOSWITCHING)

SUPPORTED UIELEMENTS SEARCH PAIRWISE COMBINATIONS

VII – RELATIVE TO SHADOW ROOT HOST COMPLEX UIELEMENT AS CLASS

59.	Page	->	ShadowRootHost_Complex_UIElement_AsClass	->	Simple_UIElement_AsInterface
60.	Page	->	ShadowRootHost_Complex_UIElement_AsClass	->	Complex_UIElement_AsClass
61.	Page	->	ShadowRootHost_Complex_UIElement_AsClass	->	UIElementsList_Of_Simple_UIElements_AsInterface
62.	Page	->	ShadowRootHost_Complex_UIElement_AsClass	->	UIElementsList_Of_Complex_UIElements_AsClass
63.	Page	->	ShadowRootHost_Complex_UIElement_AsClass	->	Frame_Simple_UIElement_AsInterface
64.	Page	->	ShadowRootHost_Complex_UIElement_AsClass	->	UIElementsList_Of_Frame_Simple_UIElements_AsInterface
65.	Page	->	ShadowRootHost_Complex_UIElement_AsClass	->	Frame_Complex_UIElement_AsClass -> Frame_Simple_UIElement_AsInterface
66.	Page	->	ShadowRootHost_Complex_UIElement_AsClass	->	ShadowRootHost_Simple_UIElement_AsInterface
67.	Page	->	ShadowRootHost_Complex_UIElement_AsClass	->	UIElementsList_Of_ShadowRootHost_Simple_UIElements_AsInterface
68.	Page	->	ShadowRootHost_Complex_UIElement_AsClass	->	ShadowRootHost_Complex_UIElement_AsClass -> ShadowRootHost_Simple_UIElement_AsInterface

VIII – RELATIVE TO UIELEMENTS LIST OF SHADOW ROOT HOST COMPLEX UIELEMENTS AS CLASS

69.	Page	->	UIElementsList_Of_ShadowRootHost_Complex_UIElements_AsClass	->	Simple_UIElement_AsInterface
70.	Page	->	UIElementsList_Of_ShadowRootHost_Complex_UIElements_AsClass	->	Complex_UIElement_AsClass
71.	Page	->	UIElementsList_Of_ShadowRootHost_Complex_UIElements_AsClass	->	UIElementsList_Of_Simple_UIElements_AsInterface
72.	Page	->	UIElementsList_Of_ShadowRootHost_Complex_UIElements_AsClass	->	UIElementsList_Of_Complex_UIElements_AsClass
73.	Page	->	UIElementsList_Of_ShadowRootHost_Complex_UIElements_AsClass	->	Frame_Simple_UIElement_AsInterface
74.	Page	->	UIElementsList_Of_ShadowRootHost_Complex_UIElements_AsClass	->	UIElementsList_Of_Frame_Simple_UIElements_AsInterface
75.	Page	->	UIElementsList_Of_ShadowRootHost_Complex_UIElements_AsClass	->	Frame_Complex_UIElement_AsClass -> Frame_Simple_UIElement_AsInterface
76.	Page	->	UIElementsList_Of_ShadowRootHost_Complex_UIElements_AsClass	->	ShadowRootHost_Simple_UIElement_AsInterface
77.	Page	->	UIElementsList_Of_ShadowRootHost_Complex_UIElements_AsClass	->	UIElementsList_Of_ShadowRootHost_Simple_UIElements_AsInterface
78.	Page	->	UIElementsList_Of_ShadowRootHost_Complex_UIElements_AsClass	->	ShadowRootHost_Complex_UIElement_AsClass -> ShadowRootHost_Simple_UIElement_AsInterface

IX – RELATIVE TO 2 SHADOW ROOT HOST COMPLEX UIELEMENTS AS CLASS

79.	Page	->	ShadowRootHost_Complex_UIElement_AsClass	->	ShadowRootHost_Complex_UIElement_AsClass	->	Simple_UIElement_AsInterface
80.	Page	->	ShadowRootHost_Complex_UIElement_AsClass	->	ShadowRootHost_Complex_UIElement_AsClass	->	Complex_UIElement_AsClass
81.	Page	->	ShadowRootHost_Complex_UIElement_AsClass	->	ShadowRootHost_Complex_UIElement_AsClass	->	UIElementsList_Of_Simple_UIElements_AsInterface
82.	Page	->	ShadowRootHost_Complex_UIElement_AsClass	->	ShadowRootHost_Complex_UIElement_AsClass	->	UIElementsList_Of_Complex_UIElements_AsClass
83.	Page	->	ShadowRootHost_Complex_UIElement_AsClass	->	ShadowRootHost_Complex_UIElement_AsClass	->	Frame_Simple_UIElement_AsInterface
84.	Page	->	ShadowRootHost_Complex_UIElement_AsClass	->	ShadowRootHost_Complex_UIElement_AsClass	->	UIElementsList_Of_Frame_Simple_UIElements_AsInterface
85.	Page	->	ShadowRootHost_Complex_UIElement_AsClass	->	ShadowRootHost_Complex_UIElement_AsClass	->	Frame_Complex_UIElement_AsClass -> Frame_Simple_UIElement_AsInterface
86.	Page	->	ShadowRootHost_Complex_UIElement_AsClass	->	ShadowRootHost_Complex_UIElement_AsClass	->	UIElementsList_Of_ShadowRootHost_Simple_UIElements_AsInterface



ACCELERATOR STATE



NON PAGE FACTORY APPROACH



DYNAMIC INSTANTIATION

```
private IUIElement? _dynamicallyInstantiatedUIElement;
1 reference 0/1 passing
public IUIElement Dynamic_IUIElement_Without_Find_Attribute_Without_Parent_Using_UIElementInstantiator
{
    get
    {
        _dynamicallyInstantiatedUIElement ??= Find<IUIElement>({
            Use.XPath,
            "//table[@id='Table2']/tr[2]/th[2]",
            name: "Dynamic_IUIElement_Without_Find_Attribute_Without_Parent_Using_UIElementInstantiator");
        return _dynamicallyInstantiatedUIElement;
    }
}
```



BASE/DEFAULT STATE



NON PAGE FACTORY APPROACH



ALL FRAMEWORKS
ARE SUPPOSED TO SUPPORT
DYNAMIC INSTANTIATION



EXEMPLAR FRAMEWORK
SUPPORT OF
DYNAMIC INSTANTIATION

```
public PayerWorkQueueGrid PayerWorkQueueGrid
=> Find<PayerWorkQueueGrid>(By.XPath("//div[@role='treegrid']"));
```




ACCELERATOR STATE



PAGE FACTORY APPROACH



STATIC INSTANTIATION

```
[Find(Use.IdEquals, "username")]  
public InputTextBox UserName;  
  
[Find(Use.ClassNameEquals, "password")]  
public InputTextBox Password;  
  
[Find(Use.DataTestIdContains, "login")]  
public IUIElement LoginButton;
```



MULTIPLE SEARCH ATTRIBUTES

```
[Find(Use.IdEquals, "multiple-find-attributes-1")]  
[Find(Use.ClassNameEquals, "multiple-find-attributes-2")]  
public IUIElementsList<IUIElement> ButtonMultipleFindAttributes;
```



BASE/DEFAULT STATE



NOT ALL FRAMEWORKS
SUPPORT PAGE FACTORY APPROACH



EXEMPLAR FRAMEWORK
SUPPORT OF
PAGE FACTORY APPROACH



EXEMPLAR FRAMEWORK
SUPPORT OF
STATIC INSTANTIATION

```
public class BasePage : WebPage  
{  
    [FindBy(How.XPath, "//*[text()='Loading']")]  
    public HtmlElement Loader { .get; .set; .}  
  
    [FindBy(How.ClassName, "app-navigation-bar")]  
    public NavigationMenu NavigationMenu { .get; .set; .}  
}
```



EXEMPLAR FRAMEWORK
DOES NOT SUPPORT
MULTIPLE SEARCH ATTRIBUTES



ACCELERATOR STATE



FRAMES SUPPORT WITH AUTOSWITCHING



FRAME ATTRIBUTE

```
[Frame(Use.IdEquals, "frame")]  
public TopLevelFrame Frame;
```



FIND ATTRIBUTE
WITH FRAME INFO

```
[Find(Use.IdEquals, "frame -> button",  
      framesFindMethod: Use.IdEquals,  
      framesFindCriteria: "frame")]  
public IUIElement ButtonInsideFrame;
```



DYNAMIC INSTANTIATION WITH FRAME INFO

```
private IUIElement? _buttonInsideFrame_DynamicInstantiation;  
public IUIElement ButtonInsideFrame_DynamicInstantiation  
{  
    get  
    {  
        if (_buttonInsideFrame_DynamicInstantiation == null)  
        {  
            UIElementsFindInfo findOption = new UIElementsFindInfo(  
                elementsFindMethod: Use.IdEquals.ToString(), elementsFindCriteria: "frame -> button",  
                framesFindMethod: Use.IdEquals.ToString(), framesFindCriteria: "frame");  
            _buttonInsideFrame_DynamicInstantiation = Find<IUIElement>(findOptions: new() { findOption },  
                name: nameof(ButtonInsideFrame_DynamicInstantiation));  
        }  
        return _buttonInsideFrame_DynamicInstantiation;  
    }  
}
```



BASE/DEFAULT STATE



MANY/MOST FRAMEWORKS

DO NOT HAVE

FRAMES SUPPORT WITH AUTOSWITCHING



EXEMPLAR FRAMEWORK

DOES NOT HAVE

FRAMES SUPPORT WITH AUTOSWITCHING



EXEMPLAR FRAMEWORK

DOES NOT SUPPORT

FIND ATTRIBUTE WITH FRAME INFO



EXEMPLAR FRAMEWORK

DOES NOT SUPPORT

FRAME ATTRIBUTES



EXEMPLAR FRAMEWORK

DOES NOT SUPPORT

DYNAMIC INSTANTIATION WITH FRAME INFO



ACCELERATOR STATE



SHADOW ROOT HOST SUPPORT WITH AUTOSWITCHING



SHADOWROOTHOST ATTRIBUTE

```
[Frame(Use.IdEquals, "shadow-root-host")]
public TopLevelShadowRootHost ShadowRootHost;
```



FIND ATTRIBUTE WITH SHADOW ROOT HOST INFO

```
[Find(Use.IdEquals, "shadow-root-host -> button",
shadowRootHostsFindMethod: Use.IdEquals,
shadowRootHostsFindCriteria: "shadow-root-host")]
public IUIElement ButtonInsideShadowRootHost;
```



DYNAMIC INSTANTIATION WITH SHADOW ROOT HOST INFO

```
private IUIElement? _buttonInsideShadowRootHost_DynamicInstantiation;

public IUIElement ButtonInsideShadowRootHost_DynamicInstantiation
{
    get
    {
        if (_buttonInsideShadowRootHost_DynamicInstantiation == null)
        {
            UIElementsFindInfo findOption = new UIElementsFindInfo(
                elementsFindMethod: Use.IdEquals.ToString(), elementsFindCriteria: "shadow-root-host -> button",
                shadowRootHostsFindMethod: Use.IdEquals.ToString(), shadowRootHostsFindCriteria: "shadow-root-host");

            _buttonInsideShadowRootHost_DynamicInstantiation = Find<IUIElement>(
                findOptions: new() { findOption },
                name: nameof(ButtonInsideShadowRootHost_DynamicInstantiation));
        }

        return _buttonInsideShadowRootHost_DynamicInstantiation;
    }
}
```



BASE/DEFAULT STATE



MANY/MOST FRAMEWORKS

DO NOT HAVE

SHADOW ROOT HOST SUPPORT WITH AUTOSWITCHING



EXEMPLAR FRAMEWORK

DOES NOT HAVE

SHADOW ROOT HOST SUPPORT WITH AUTOSWITCHING



EXEMPLAR FRAMEWORK

DOES NOT SUPPORT

FIND ATTRIBUTE WITH SHADOW ROOT HOST INFO



EXEMPLAR FRAMEWORK

DOES NOT SUPPORT

SHADOW ROOT HOST ATTRIBUTES



EXEMPLAR FRAMEWORK

DOES NOT SUPPORT

DYNAMIC INSTANTIATION WITH SHADOW ROOT HOST INFO



ACCELERATOR STATE



SUPPORT OF COMPLEX SCENARIOS WITH AUTOSWITCHING



UIELEMENT INSIDE FRAME WITHIN SHADOW ROOT HOST USING ATTRIBUTE

```
[Find(shadowRootHostsFindMethod: Use.IdEquals, shadowRootHostsFindCriteria: "shadow-root-host",
framesFindMethod: Use.IdEquals, framesFindCriteria: "shadow-root-host -> frame",
elementsFindMethod: Use.IdEquals, elementsFindCriteria: "shadow-root-host -> frame -> button",
findOrderType: FindOrderType.FrameInsideShadowRootHost)]
public IUIElement ButtonInsideFrameWithinShadowRootHost;
```



FRAME INSIDE SHADOW ROOT HOST USING ATTRIBUTE

```
[Frame(framesFindMethod: Use.IdEquals,
framesFindCriteria: "shadow-root-host -> frame",
shadowRootHostsFindMethod: Use.IdEquals,
shadowRootHostsFindCriteria: "shadow-root-host")]
public FrameInsideShadowRootHost FrameInsideShadowRootHost;
```



SHADOW ROOT HOST INSIDE FRAME USING ATTRIBUTE

```
[ShadowRootHost(shadowRootHostsFindMethod: Use.IdEquals,
shadowRootHostsFindCriteria: "frame -> shadow-root-host",
framesFindMethod: Use.IdEquals,
framesFindCriteria: "frame")]
public ShadowRootHostInsideFrame ShadowRootHostInsideFrame;
```



UIELEMENT INSIDE SHADOW ROOT HOST WITHIN FRAME USING DYNAMIC INSTANTIATION

```
private IUIElement? _buttonInsideShadowRootHostWithinFrame_DynamicInstantiation;
public IUIElement ButtonInsideShadowRootHostWithinFrame_DynamicInstantiation
{
    get
    {
        if (_buttonInsideShadowRootHostWithinFrame_DynamicInstantiation == null)
        {
            UIElementsFindInfo findOption = new UIElementsFindInfo(
                framesFindMethod: Use.IdEquals.ToString(), framesFindCriteria: "frame",
                shadowRootHostsFindMethod: Use.IdEquals.ToString(), shadowRootHostsFindCriteria: "frame -> shadow-root-host",
                elementsFindMethod: Use.IdEquals.ToString(), elementsFindCriteria: "frame -> shadow-root-host -> button",
                findOrderType: FindOrderType.ShadowRootHostInsideFrame);

            _buttonInsideShadowRootHostWithinFrame_DynamicInstantiation = Find<IUIElement>(
                findOptions: new() { findOption },
                name: nameof(ButtonInsideShadowRootHostWithinFrame_DynamicInstantiation));
        }

        return _buttonInsideShadowRootHostWithinFrame_DynamicInstantiation;
    }
}
```



BASE/DEFAULT STATE



MANY/MOST FRAMEWORKS

DO NOT HAVE

SUPPORT OF COMPLEX SCENARIOS WITH AUTOSWITCHING



EXEMPLAR FRAMEWORK

DOES NOT HAVE

SUPPORT OF COMPLEX SCENARIOS WITH AUTOSWITCHING



EXEMPLAR FRAMEWORK

DOES NOT SUPPORT

UIELEMENT INSIDE FRAME WITHIN SHADOW ROOT HOST SING ATTRIBUTE



EXEMPLAR FRAMEWORK

DOES NOT SUPPORT

FRAME INSIDE SHADOW ROOT HOST USING ATTRIBUTE



EXEMPLAR FRAMEWORK

DOES NOT SUPPORT SCENARIO

SHADOW ROOT HOST INSIDE FRAME USING ATTRIBUTE



EXEMPLAR FRAMEWORK

DOES NOT SUPPORT SCENARIO

UIELEMENT INSIDE SHADOW ROOT HOST WITHIN FRAME USING DYNAMIC INSTANTIATION



ACCELERATOR STATE

SUPPORTED SEARCH UIELEMENTS METHODS

namespace DevQAProdCom.Core.UI.Shared.Enumerations

```

public enum Use //taken from Playwright
{
    NotSet,
    //taken from Selenium
    IdEquals,
    IdContains,
    NameEquals,
    NameContains,
    ClassNameEquals,
    ClassNameContains,
    LinkTextEquals,
    LinkTextContains,
    TagName,
    CssSelector,
    XPath,
    TextEquals,
    TextContains,
    LabelEquals,
    LabelContains,
    PlaceholderEquals,
    PlaceholderContains,
    AltTextEquals,
    AltTextContains,
    TitleEquals,
    TitleContains,
    DataTestIdEquals,
    DataTestIdContains,
}

```



CUSTOM SEARCH UIELEMENTS METHODS

```

_serviceCollection.AddPlaywrightFindOptionSearchMethod<PlaywrightCustomFindOptionSearchMethod>(C);

public class PlaywrightCustomFindOptionSearchMethod()
    : BasePlaywrightFindOptionSearchMethodWithCustomSelector
{
    4 references
    public override string Method => Const.Ui.CustomFindOptionSearchMethod;
    7 references
    protected override string GetSelector(IFindOption findOption)
        => string.Concat("css=", $"{Const.Ui.AttributeForCustomFindOptionSearchMethod}='{findOption.Criteria}'");
}

```



BASE/DEFAULT STATE

EXEMPLAR FRAMEWORK SUPPORTED SEARCH UIELEMENTS METHODS

```

public enum How
{
    XPath,
    CssSelector,
    Id,
    TagName,
    ClassName,
    LinkText,
    Name,
    PartialLinkText,
    jQuery
}

```

CUSTOM SEARCH UIELEMENTS METHODS:



NOT ALL FRAMEWORKS
SUPPORT CUSTOM SEARCH UIELEMENTS METHODS



EXEMPLAR FRAMEWORK
SUPPORTS
CUSTOM SEARCH UIELEMENTS METHODS

```

using IJavaScriptExecutor = IJavaScriptExecutor;

2 references | 0 changes | 0 authors, 0 changes
internal class ByjQuery : OpenQA.Selenium.By
{
    private readonly string _selector;

    1 reference | 0 changes | 0 authors, 0 changes
    public ByjQuery(string selector)
    {
        _selector = $"{selector}";
    }

    0 references | 0 changes | 0 authors, 0 changes
    public override IWebElement FindElement(ISearchContext context)
    {
        var element = GetJavaScriptObject(context, 0);
        if (element is IWebElement) return (IWebElement)element;
        throw new NoSuchElementException($"No element found with jQuery command: {jQuery} + {_selector}");
    }
}

```

ADD SLIDE WITH BEHAVIORS

LOGGING





ACCELERATOR STATE



ADVANCED PARAMETERIZED LOGGING WITH NUNIT ADDIN



NOT HARDCODED - INTERFACE BASED
SERILOG LIBRARY IS USED AS DEFAULT
BUT CAN BE SUBSTITUTED TO ANY OTHER
WITHOUT CODE BASE CHANGE



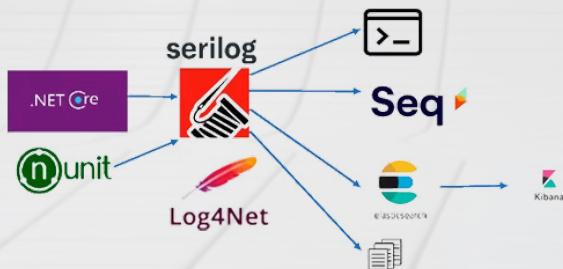
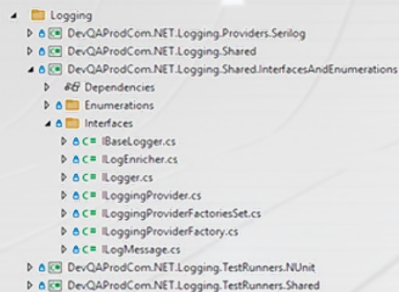
SEVERAL LIBRARIES CAN BE USED
SERILOG AND LOG4NET, AND ...

USE CASE 1:

TECHNOLOGY BECAME PAID OR NOT SUPPORTED
SMOOTH TRANSITION BETWEEN TECHNOLOGIES
TEST FIRST ANY NEW IN PARALLEL -> THEN MAKE TRANSITION

USE CASE 2:

NEED TO LOG TO 2 DESTINATIONS AT THE SAME TIME
LIBRARY 1 - SUPPORTS DESTINATION 1
LIBRARY 2 - SUPPORTS DESTINATION 2



BASE/DEFAULT STATE



SUPPOSEDLY NONE OF FRAMEWORKS
SUPPORTS ADVANCED PARAMETERIZED LOGGING
EXEMPLAR FRAMEWORK DOES NOT SUPPORT



MOST FRAMEWORKS USE EITHER :
- NON-PARAMETERIZED LOGGING
- STANDARD (NOT ADVANCED) PARAMETERIZED LOGGING



HARDCODED SERILOG/ANY OTHER LIBRARY CLASSES
REQUIRE ALL LOG CODE CHANGE IN CASE OF SUBSTITUTION



SINGLE LIBRARY USAGE SUPPORT

```
public IEnumerable<GetRoleDto> GetAllowedRolesByUserId(string userId)
{
    Log.Information("Get-user-roles-by-user-ID: {0}", userId);
    // ...
}
// class Serilog.Log
// An optional static entry point for logging that can be easily referenced by different parts of
```





ACCELERATOR STATE



**SPECIALIZED LOGGING MODELS
FOR ADVANCED PARAMETERIZED LOGGING**



BASE/DEFAULT STATE



**MOST/ALMOST ALL FRAMEWORKS
DOES NOT SUPPORT
SPECIALIZED LOGGING MODELS**



**EXEMPLAR FRAMEWORK
DOES NOT SUPPORT
SPECIALIZED LOGGING MODELS**

Message:

System.TimeoutException : Element TextBox. XPath, ../input[@autocomplete='username_field'] is not displayed after timeout

```
Should_Support_Search_Page_FrameComplexUIElementAsClass_FrameComplexUIElementAsClass_ShadowRootHostComplexUIElementAsClass_ShadowRootHostSimpleUIElementAsInterface
Source: Tests_UIElement_Search_RelativeTo_2FramesComplexUIElementsAsClass.cs line 178
Duration: 15,2 sec

Message:
DevQAProdCom.NET.UI.Shared.OperativeClasses.UIElements.UIElementNotFoundException : No native elements were found or errors appeared while searching for UIElement 'Page -> Page_FrameComplexUIElementAsClass_29f4 ->
FrameComplexUIElementAsClass_f2ab -> ShadowRootHostComplexUIElementAsClass_f2ab -> ShadowRootHostSimpleUIElementAsInterface_f2ab' on page 'http://localhost:44497/TestPage_UIElementsSearchRelativeTo2FramesComplexUIElementsAsClass'.
{
  "SearchResult": {
    "UriAfterSearch": "http://localhost:44497/TestPage_UIElementsSearchRelativeTo2FramesComplexUIElementsAsClass",
    "UIElementFullName": "Page -> Page_FrameComplexUIElementAsClass_29f4 -> FrameComplexUIElementAsClass_f2ab -> ShadowRootHostComplexUIElementAsClass_f2ab -> ShadowRootHostSimpleUIElementAsInterface_f2ab",
    "UIElementName": "ShadowRootHostSimpleUIElementAsInterface_f2ab",
    "GeneralFindState": "Failed",
    "GeneralExceptionMessage": "No native elements were found or errors appeared while searching for UIElement 'Page -> Page_FrameComplexUIElementAsClass_29f4 -> FrameComplexUIElementAsClass_f2ab ->
ShadowRootHostComplexUIElementAsClass_f2ab -> ShadowRootHostSimpleUIElementAsInterface_f2ab' on page 'http://localhost:44497/TestPage_UIElementsSearchRelativeTo2FramesComplexUIElementsAsClass'.",
    "TotalAmountOfFindChainsProcessed": 1,
    "FindChains": {
      "Found": {},
      "Not Found": {},
      "Failed": {
        "Chain 0": [
          {
            "NestingLevel": 0,
            "FindState": "Failed",
            "Name": "Page_FrameComplexUIElementAsClass_29f4",
            "UiIndex": null,
            "IsElementOfList": false,
            "FindInfo": {
              "UIElementType": "Frame",
              "SearchKind": "FrameElements",
              "FindOrderType": "NotSet",
              "ElementsFindOption": null,
              "FramesFindOption": {
                "Method": "IdEquals",
                "Criteria": "page->frameComplexUIElementAsClass(29f4)"
              },
              "ShadowRootHostsFindOption": null
            },
            "TotalAmountOfElementsFound": 0,
            "ExceptionMessage": "Single native element was expected, but 0 native elements were found while searching for find chain UI element 'Page_FrameComplexUIElementAsClass_29f4' using method and criteria: 'Name:
Page_FrameComplexUIElementAsClass_29f4'. FindParameters: <Frames: Use.IdEquals('page->frameComplexUIElementAsClass(29f4)')>'.",
            "StackTrace": " at DevQAProdCom.NET.UI.Shared.OperativeClasses.UIElements.Search.BaseNativeElementsSearcher`3.GetFindParametersResultsList[T](List`1 elements, IFindParametersWithSearchResult
```


[illegible]

SELECT ALL PAGES TESTED (in SPECIFIED RUN)

Seq Personal 

```
select distinct(PageAbsoluteUri) as AllPagesTested
from stream
where Operation = 'UiElementSearch'
and TestRunName = 'RunName: RegressionTestRun' and TestRunId = 'RunId: 2365a'
order by AllPagesTested
```



AllPagesTested
http://localhost:44497/DragAndDropTestPage
http://localhost:44497/FramesTestPage
http://localhost:44497/HomeTestPage
http://localhost:44497/KeyboardTestPage
http://localhost:44497/LoginTestPage
http://localhost:44497/ScrollTestPage
http://localhost:44497/SomePath/TestPageTab1
http://localhost:44497/SomePath/TestPageTab2
http://localhost:44497/SomePath2/TestPageTab3
http://localhost:44497/TestPage2

Computed in 66.8 ms

SELECT ALL COMPONENTS TESTED (on SPECIFIED PAGE)

Seq  Personal 	
<pre>select distinct(UiElementName) as UiElementName from stream where PageAbsoluteUri = 'http://localhost:44497/TestPage2' and TestRunName = 'RunName: RegressionTestRun' and TestRunId = 'RunId: 2365a' group by PageAbsoluteUri order by PageAbsoluteUri</pre>	
  	
 Last	
PageAbsoluteUri	UiElementName
http://localhost:44497/TestPage2	MouseUpButton
http://localhost:44497/TestPage2	UseLinkTextContains
http://localhost:44497/TestPage2	Dynamic_UiElementsList_IUIElement_Without_Find_Attribute_With_Parent_Using_UiElementInstantiator
http://localhost:44497/TestPage2	UsePlaceholderEquals
http://localhost:44497/TestPage2	UseTitleContains
http://localhost:44497/TestPage2	DeleteTableButton
http://localhost:44497/TestPage2	Form
http://localhost:44497/TestPage2	Dynamic_IUIElement_Without_Find_Attribute_With_Parent_Using_UiElementInstantiator
http://localhost:44497/TestPage2	Rows3
http://localhost:44497/TestPage2	NotExistingInTheDomList
http://localhost:44497/TestPage2	MouseClickedDoubleClickButton
http://localhost:44497/TestPage2	UseCustomFindOptionSearchMethodRegisteredFromDi
http://localhost:44497/TestPage2	UList
http://localhost:44497/TestPage2	UseTitleEquals
http://localhost:44497/TestPage2	Rows
http://localhost:44497/TestPage2	UseXPath
http://localhost:44497/TestPage2	UseAltTextContains
http://localhost:44497/TestPage2	UseTextContains
http://localhost:44497/TestPage2	MouseHoverButton
http://localhost:44497/TestPage2	TextArea
http://localhost:44497/TestPage2	ElementForPageMouseMove
http://localhost:44497/TestPage2	UseAltTextEquals

SELECT ALL TESTS which TEST SPECIFIED PAGE

Seq

Personal

```
select distinct(TestMethodName) as TestMethodName
from stream
where PageAbsoluteUri = 'http://localhost:44497/TestPage2'
and TestRunName = 'RunName: RegressionTestRun' and TestRunId = 'RunId: 2365a'
group by PageAbsoluteUri
order by PageAbsoluteUri
```



PageAbsoluteUri	TestMethodName
http://localhost:44497/TestPage2	Should_MouseClick_UIElement
http://localhost:44497/TestPage2	Should_TextArea_Handle_SetText_And_GetInputText_Behaviors
http://localhost:44497/TestPage2	Should_Get_Dynamic_UIElementsList_UIElement_Without_Find_Attribute_Without_Parent_Using_UIElementInstantiator
http://localhost:44497/TestPage2	Should_Get_Dynamic_UIElementsList_TUIElement_Without_Find_Attribute_Without_Parent_Using_UIElementInstantiator
http://localhost:44497/TestPage2	Should_Find_UIElement_Using_Text_Equals
http://localhost:44497/TestPage2	Should_Get_Table3_Rows1_Cells1_Ullist_Select_Text
http://localhost:44497/TestPage2	Should_Return_Primitive_Value_Using_Execute_Async_JavaScript_T_Relative_To_UIElement
http://localhost:44497/TestPage2	Should_Find_UIElement_Using_Placeholder_Equals
http://localhost:44497/TestPage2	Should_Get_Dynamic_UIElement_Without_Find_Attribute_With_Parent_Using_UIElementInstantiator
http://localhost:44497/TestPage2	Should_Return_Non_Primitive_Value_Using_Execute_JavaScript_T_Relative_To_Page
http://localhost:44497/TestPage2	Should_Find_UIElement_Using_Custom_Find_Option_Search_Method_Registered_From_Activator_Create_Instance_T
http://localhost:44497/TestPage2	Should_Return_Non_Primitive_Value_Using_Execute_Async_JavaScript_T_Relative_To_UIElement
http://localhost:44497/TestPage2	Should_Get_Table3_Rows1_Cells1_Ullist_Select_Text_FOREACH_WithEnumerator
http://localhost:44497/TestPage2	Should_Find_UIElement_Using_Placeholder_Contains
http://localhost:44497/TestPage2	Should_Execute_Void_Async_JavaScript_Relative_To_UIElement_As_Parent_Container
http://localhost:44497/TestPage2	Should_Find_UIElement_Using_Text_Contains
http://localhost:44497/TestPage2	Should_Return_Non_Primitive_Value_Using_Execute_Async_JavaScript_T_Relative_To_UIElement_As_Parent_Container
http://localhost:44497/TestPage2	Should_Find_UIElement_Using_IdEquals
http://localhost:44497/TestPage2	Should_Find_UIElement_Using_AltText_Contains
http://localhost:44497/TestPage2	Should_Find_UIElement_Using_TagName
http://localhost:44497/TestPage2	Should_Execute_Void_JavaScript_Relative_To_Page
http://localhost:44497/TestPage2	Should_Get_Table3_Rows1_Cells1_Ullist_Select_Text_FOREACH
http://localhost:44497/TestPage2	Should_Get_Dynamic_UIElement_Without_Find_Attribute_Without_Parent_Using_UIElementInstantiator
http://localhost:44497/TestPage2	Should_Get_Table2_Rows_Select_Cells_Text
http://localhost:44497/TestPage2	Should_Selenium_UIElement_Contain_Native_Objects
http://localhost:44497/TestPage2	Should_InputTextBox_Handle_SetText_And_GetInputText_Behaviors

SELECT ALL TESTS which TEST SPECIFIED COMPONENT on SPECIFIED PAGE

Seq  Personal 

```
select distinct(TestMethodName) as TestMethodName
from stream
where PageAbsoluteUri = 'http://localhost:44497/TestPage2'
and UiElementName = 'Form'
and TestRunName = 'RunName: RegressionTestRun' and TestRunId = 'RunId: 2365a'
group by PageAbsoluteUri, UiElementName
order by PageAbsoluteUri
```

PageAbsoluteUri	UiElementName	TestMethodName
http://localhost:44497/TestPage2	Form	Should_Execute_Void_JavaScript_Relative_To_UiElement_As_Parent_Container
http://localhost:44497/TestPage2	Form	Should_Return_Primitive_Value_Using_Execute_Async_JavaScript_T_Relative_To_UiElement_As_Parent_Container
http://localhost:44497/TestPage2	Form	Should_Execute_Void_Async_JavaScript_Relative_To_UiElement_As_Parent_Container
http://localhost:44497/TestPage2	Form	Should_Return_Non_Primitive_Value_Using_Execute_JavaScript_T_Relative_To_UiElement_As_Parent_Container
http://localhost:44497/TestPage2	Form	Should_Return_Non_Primitive_Value_Using_Execute_Async_JavaScript_T_Relative_To_UiElement_As_Parent_Container
http://localhost:44497/TestPage2	Form	Should_Return_Primitive_Value_Using_Execute_JavaScript_T_Relative_To_UiElement_As_Parent_Container

Computed in 13.9 ms

OTHER QUERIES OF ANY COMPLEXITY

- **SELECT TESTS BY CATEGORIES**
- **SELECT PASSED/FAILED TESTS**
- **SELECT API ENDPOINTS COVERED BY TESTS**
- **ETC.**

USAGE AREAS:

- **BUILD DASHBOARDS with REGRESSION TESTING RESULTS**
- **BUILD ANY OTHER TYPES OF TEST REPORTS based on COLLECTED DATA**

The background image is a grayscale, high-tech control room or bridge. It features multiple consoles with various screens and buttons. The central console has a large screen displaying a complex network diagram. The room has a futuristic, industrial feel with various panels and lights. The word "CONFIGURATIONS" is overlaid in the center in a bold, black, sans-serif font.

CONFIGURATIONS



ACCELERATOR STATE



NO NEED TO HARDCODE NAMES OF CONFIG FILES
AUTOMATIC PROCESSING OF JSON, XML, INI
FILES FROM FOLDERS



AUTOMATIC PROCESSING FROM:

- DEFAULT «CONFIGURATIONS» FOLDER
- «CONFIGURATIONS\SHARED» FOLDER
- «CONFIGURATIONS\{ENV}» FOLDER
- ANY SPECIFIED FOLDER PATH

```
Tests.DevQAProdCom.NET.Configurations
├── Dependencies
├── Configurations
│   ├── DEV
│   │   ├── appsettings.Tests.DEV.ini
│   │   ├── appsettings.Tests.DEV.json
│   │   ├── appsettings.Tests.DEV.xml
│   │   ├── ignore_appsettings.Tests.DEV.json
│   │   └── testData.DEV.json
│   ├── QA
│   │   ├── appsettings.Tests.QA.ini
│   │   ├── appsettings.Tests.QA.json
│   │   ├── appsettings.Tests.QA.xml
│   │   └── testData.QA.json
│   └── Shared
│       ├── sharedConfigurations.json
│       ├── environment.json
│       └── GlobalConfigContainer.cs
```



BASE/DEFAULT STATE

MOST FRAMEWORKS (so as EXEMPLAR FRAMEWORK):



HARDCODED NAMES OF CONFIG FILES



NO AUTOMATIC PROCESSING

```
public static class Globals
{
    // This is the base configuration builder that loads the common non-secret appsettings files
    // 4 references | Vladyslav Babarkib, 2017 days ago | 1 author, 1 change
    private static IConfigurationBuilder GetBaseConfigurationBuilder() => new ConfigurationBuilder()
        .AddJsonFile(NonSecretAppSettings)
        .AddJsonFile("appsettings.Shared.json")
        .AddJsonFile("appsettings.local.json", optional: true)
        .AddEnvironmentVariables(); // PRESERVE ORDER. ENVIRONMENT VARIABLES SHOULD BE AT THE END,
        // CAUSE THEY OVERRIDE SOME VALUES FROM APPSETTINGS.JSON FILE IN CI/CD.

    private static string NonSecretAppSettings =>
        $"appsettings.{Const.Config.Applications.App}.{{TestEnv.App Environment.ToString().ToLower()}}.json";
}
```